



Downtown of Beirut, Lebanon by Night



Where expertise meets innovation

Company Profile

2025

Who We Are



We are established in 2008, specializing in the realm of Mechanical, Electrical, Plumbing, ICT and Renewable Energy Technologies, covering the spectrum of Design, Shop Drawings and Value Engineering services offerings.

We are guided by optimism, using the power of design to inspire a positive change and create a future that promotes wellbeing for all, in line with the latest sustainability trends. We focus on the importance of preserving nature and its resources and encourage our clients to apply any opportunity to reduce the overall impact of the built-up environment on the human health and the natural ecosystem.

We are a diverse team, composed of experts in the latest Engineering trends in Renewable Energies, Green Building designs, HVAC Systems technologies, Water Systems efficiencies, Power Generation & Distribution as well as new technologies in IT & Low Current systems. Our aim is to provide Total MEP Integrated Solutions, customized to meet the full spectrum of Engineering services through a constructive cross-discipline collaboration.

We invest in innovation, meeting market's expectations, by pushing the boundaries of design through smartly using AEC modeling technologies offering integrated construction models and virtual reality rendering. We are unlocking and creating new tools to release fast innovative solutions serving the construction industry. We promote education within our teams specially in terms of technology and sustainability, through certificates and diplomas acquired from accredited universities and international organizations.

We expand our market, by benefiting from the communications technology evolutions, to connect with new clients expanding from the Gulf Countries to North American Countries - mainly Canada. From our side, this expansion enabled us to gain a competitive edge in undersaturated markets, foster our expertise through the diversity of shared opinions and methodologies; from our client's side, this expansion offered them the opportunity to diversify their services, to access additional talents and reduce costs.

We come towards our clients, by creating CONSULT EMDC in Dubai, UAE to become an attraction hub for an expanding clientele. The scope of CONSULT EMDC is to proactively recognize and pursue opportunities, cultivate old and new clients' relationships achieving long-term sustainability in the MEP Construction and manage the Business Development, Contracts, Sales and Purchasing activities of both EMDC Group and CONSULT EMDC.

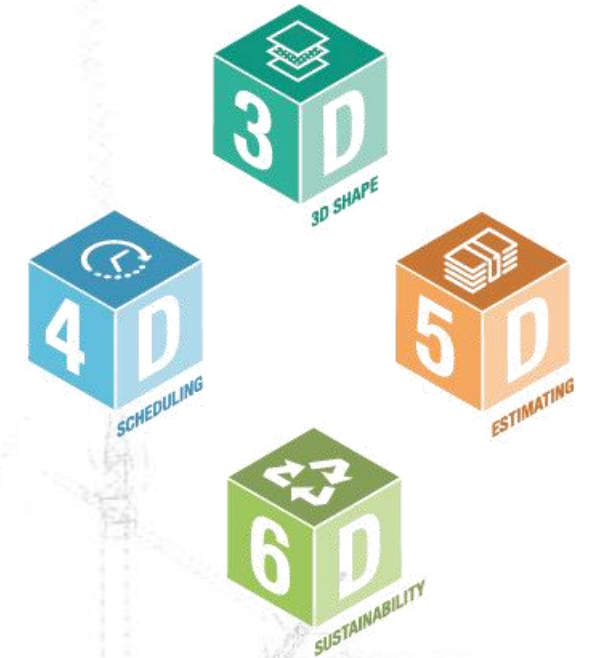
Our Projects Locations



The use of BIM is now mandated to ascertain that the planning, design, and construction of projects is a collaborative process with a high-efficiency output. For this purpose, EMDC invested in the BIM world allowing the 3D simulation of the projects before their execution, facilitating visualization and coordination, analyzing and resolving potential clashes, reducing the risk of abortive works on site and finally making more informed decisions.

Further to the 3D modeling employed for construction and fabrication, EMDC can offer its clients the essential functionalities of BIM models through BIM 4D to 6D, starting with scheduling tasks, estimating costs and extracting quantities and performing energy analysis.

Throughout the projects, EMDC got acquainted with BIM components which play a crucial role in the successful implementation and management of BIM projects, like Common Data Environment (CDE), BIM Execution Plan (BEP), COBie (Construction Operations Building Information Exchange), IFC (Industry Foundation Classes), Asset Information Model (AIM), FM (Facility Management) Integration, GIS (Geographic Information Systems) Integration, 4D and 5D BIM, Virtual Reality (VR) & Augmented Reality (AR) as well as Smart Building Technologies.



We, at EMDC, enriched our BIM expertise by closely developing BIM models to renown companies, listed in a later section, integrating the MEP Revit materials families from their selected suppliers, their method of statement and showing their installation and fabrication methods in their details and finally submitting a coordinated model as per the presentation technique expected in each company.

Finally, as a contribution to the worldwide BIM community, we participated in Autodesk University classes to share our insights about the Automation tools developed in house, automating time-consuming design tasks using Forge and Machine Learning. Both classes are found at the following links:

<https://www.autodesk.com/autodesk-university/article/Combining-Forge-and-Machine-Learning-Automate-Time-Consuming-Design-Tasks-2021>

<https://www.autodesk.com/autodesk-university/class/Beyond-Dynamo-Powerful-Automation-Potential-Forge-and-Revit-API-2021>



Design expertise

We gain our ample expertise through close coordination with International Designers and while executing Value Engineering and Shop Drawings projects.

Shop Drawings

We delivered 10,000 shop drawings within 9 months including client's comment implementation as well as a LOD 400 Revit model and Navisworks model with zero clash for a 250,000 sq.m. mega carpark within 4 months. Moreover, we developed fabrication drawings for projects in Canada, USA and Australia.

Value Engineering & Design Review

We performed Value Engineering and Design Review tasks on designs made by renown international companies.

BIM Proficiency

We delivered coordinated Revit models using advanced BIM logistics like Navisworks clash tests, MagiCAD and plugins and add-ons. We continuously keep our teams updated with the latest changes of Autodesk AEC collection.

Development of Virtual Reality & Augmented Reality

We provided our clients with a collection of navigation tools by converting selected areas with particular importance into virtual and interactive environments.

Data Security

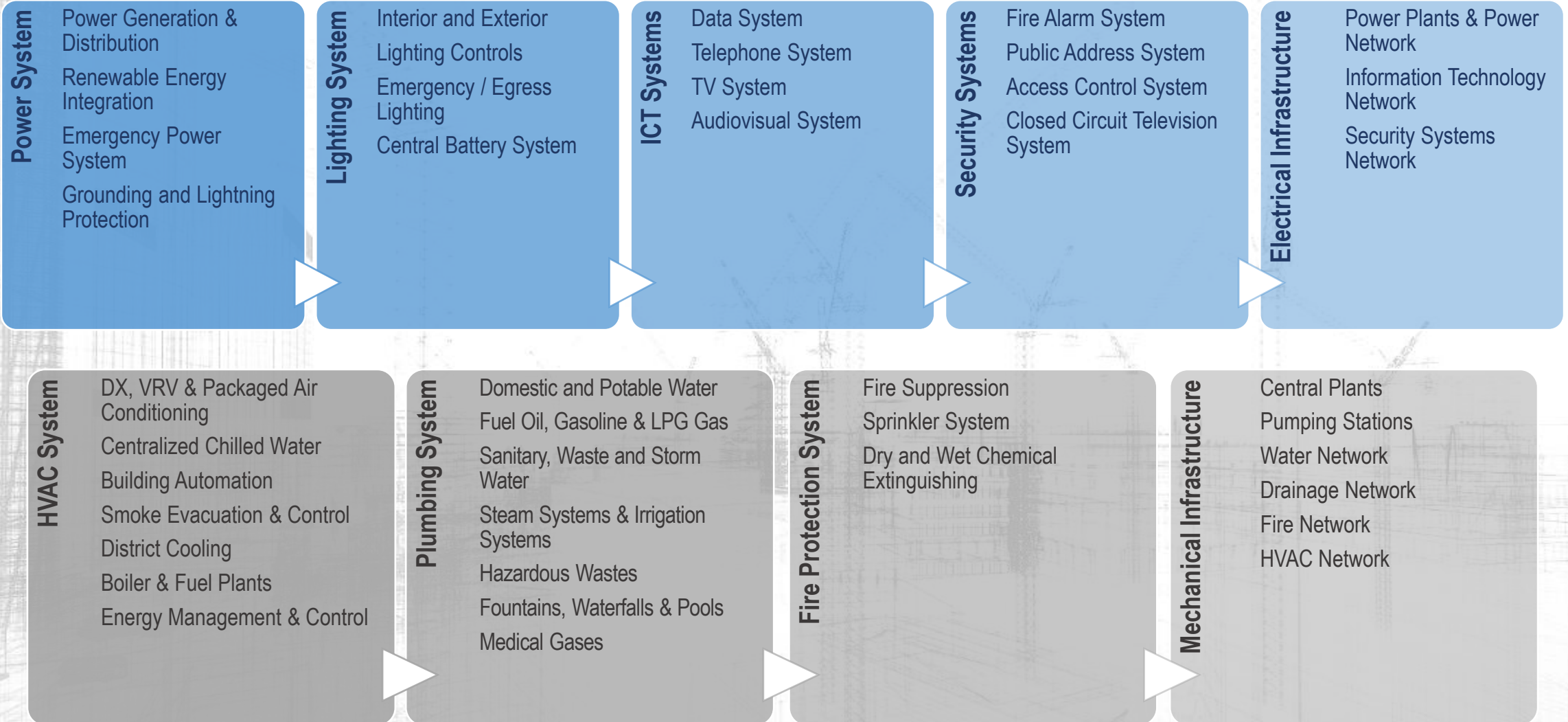
We securely back-up our projects on redundant servers with redundant internet connections as well as via subscription on a Remote Server (Cloud), through latest HP servers Technology.

In House R&D

We internally developed automation processes and even featured them in Autodesk University.

Also, we developed in-house BIM tools related to each project's needs.

What We Do : MEP & ICT



Renewable Energy Systems

Solar Poly-Generation

- Solar Cooling
- Solar Heating
- Solar for Hot Water Generation
- 100% Solar Fraction System
- Thermodynamic Solar System
- PV-Heat Pump System
- Organic Rankine Cycle
- Phase Change Thermal Storages

Solar Cooling

PV Electrical Power Generation

Smart Thermal Networks

Smart Grid

District Solar Cooling/ Heating Plants

Geothermal Energy

Sustainability & Green Design

Environmentally responsible and resource-efficient Design

Efficient use of Energy, Water and other Resources

Protection of Occupants' Health

Improvement of Productivity

Reduction of Waste and Pollution

Energy Modeling

Energy Solutions

Energy Management Analysis

Energy Reduction Percentage from systems enhancements

Optimizing efficient HVAC Systems

Full Energy Model for Cooling, Heating, Hot Water Energy

Methodologies towards Near-Zero Energy Buildings (nZEBs)

Energy Audit

Pre-Site Stage

Document Collection Stage

Site Stage

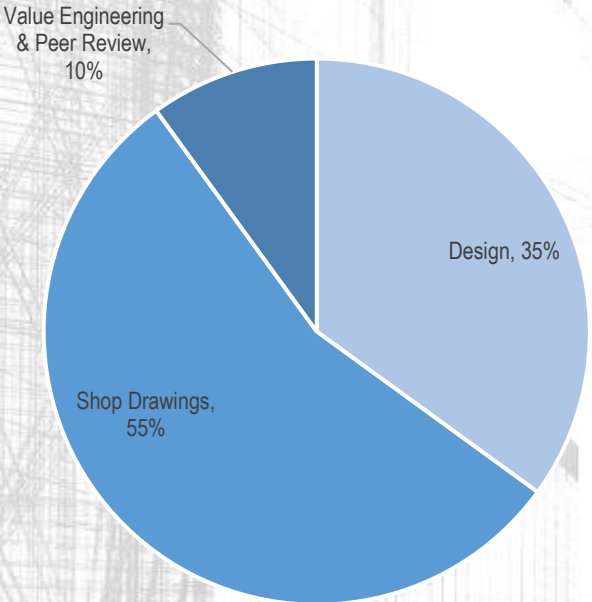
Basic Understanding

Post Site Stage

Evaluation Stage with Final Report

We, at EMDC, acquired the essential diplomas and certifications in Energy & Sustainability.

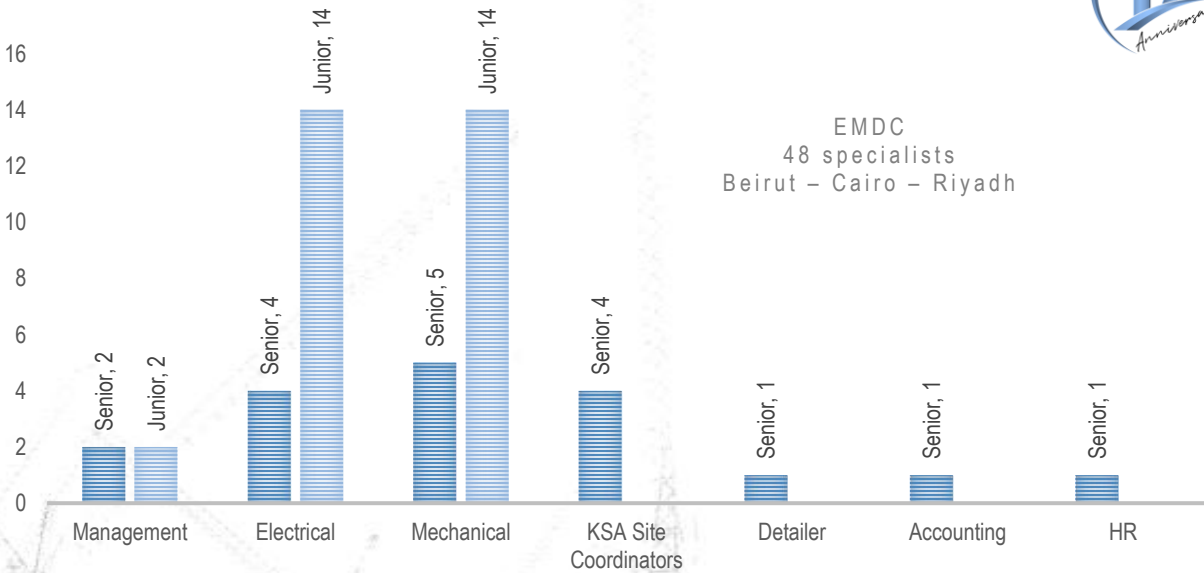
Our Projects & Workforce



- Design
- Airports & Stations
 - Commercial
 - Cultural
 - Education
 - Exhibitions
 - Governmental
 - Healthcare
 - High Rise
 - Hospitality
 - Industrial
 - Infrastructure
 - Leisure
 - Offices
 - Plants
 - Religious
 - Residential

- Shop Drawings
- Airports & Stations
 - Commercial
 - Healthcare
 - High Rise
 - Hospitality
 - Industrial
 - Infrastructure
 - Leisure
 - Offices
 - Plants
 - Residential

- Value Engineering
- Airports & Stations
 - Commercial
 - Governmental
 - Healthcare
 - High Rise
 - Hospitality
 - Industrial
 - Offices



At EMDC, we specialize in BIM modeling and coordination, providing professional expertise in implementing the BIM platform across various projects. From the outset of each project, a dedicated BIM Manager is assigned to ensure that the BIM methodology is fully understood and seamlessly integrated. This ensures that the project team can effectively apply BIM principles throughout the project's lifecycle, optimizing efficiency, collaboration, and precision.

Our Major Clients



SAMSUNG C&T



dar al-handasah
shair and partners



Rafik El-Khoury & Partners
Consulting - Engineers



INTERNATIONAL
L'INGÉNIEUR DE LA CONSTRUCTION



FIRST GULF COMPANY
الشركة الخليجية الأولى



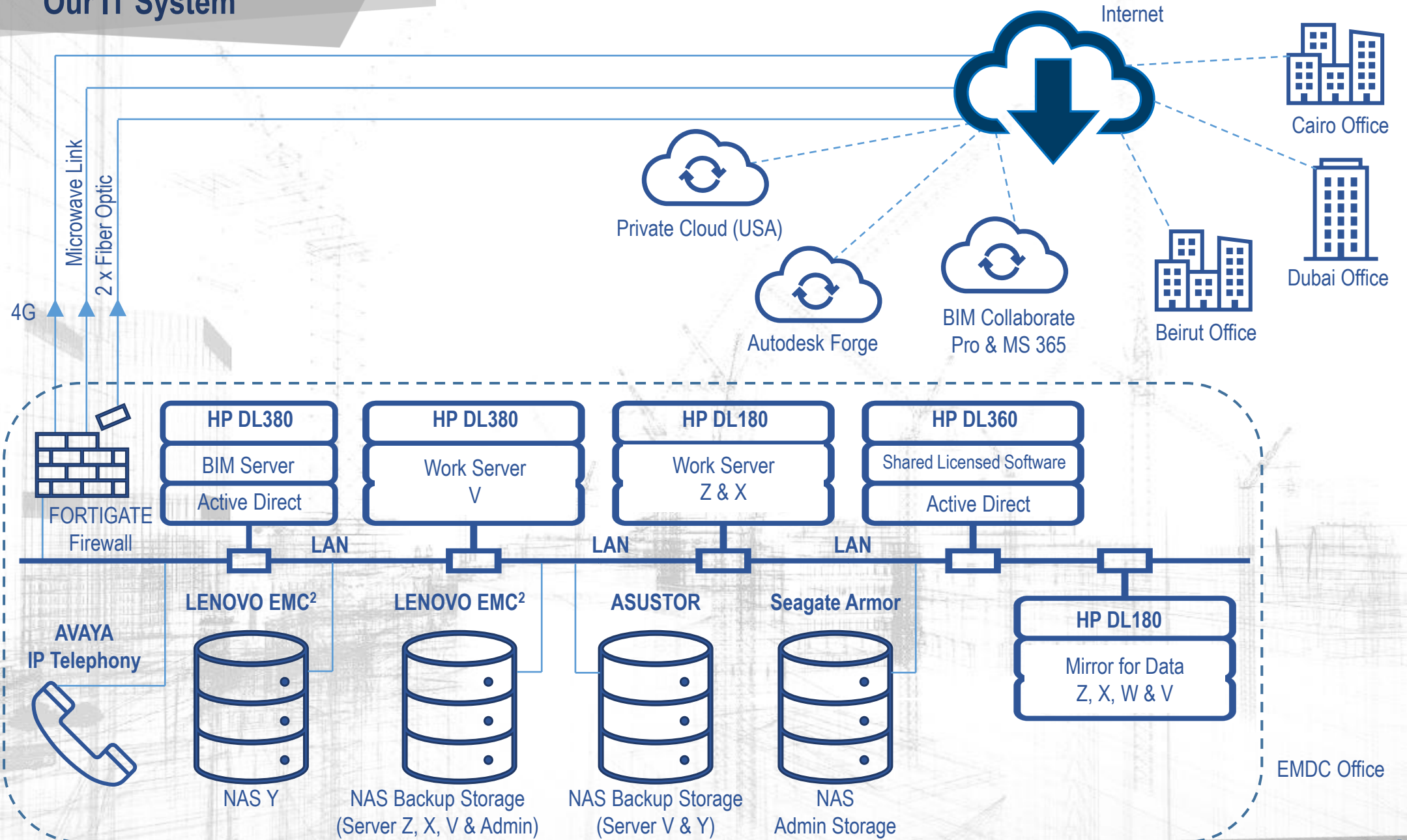
مجموعة بن لادن السعودية
SAUDI BINLADIN GROUP



I.T.B. Nigeria Limited



Our IT System





1987



2013



2014



2017



2019

Codes & Standards. Relevant codes set minimum standards for the design and construction of projects, including requirements for fire and life safety, accessibility and constructability.

Our team is well-informed about all recognized standards such as American Society of Heating, Refrigerating & Air-Conditioning Engineers (ASHRAE), British Standards (BS), DEWA Regulations for Electrical Installations, European Norms (EN), Gulf Countries Codes (GCC), International Building Code (IBC), National Fire Protection Association (NFPA), Normes Françaises (NF), Saudi Building Code (SBC), Telecommunications Industry Association (TIA-EIA), etc. and the like.

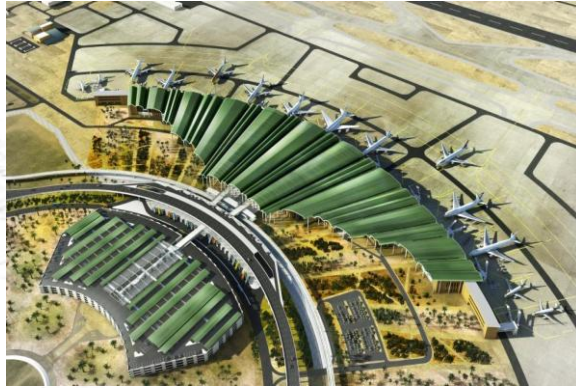
Software. Calculations and simulation software allow MEP engineers to evaluate the performance of their systems prior to implementation in the design. These tools help equipments arrangements, sizing and compliance to code requirements. Our team is always ahead of the latest software covering systems like Lighting, Power, Photovoltaic, Telecom, Sound, CCTV, Grounding & Lightning as well as Firefighting, HVAC Load Analysis, Ducting, CFD Simulation, Smoke Management and Plumbing.

Drawing & Modeling. CAD drawing ensures the ability to provide detailed and accurate representations of designs, allowing professionals to visualize the design in a virtual environment, and making necessary changes and modifications before submission to tender. In addition, 3D modeling enables a greater coordination between different design teams and disciplines, which helps to avoid errors and conflicts in the construction phase, prior to physical implementation. Our team has a high level of proficiency in the renown AutoCAD, REVIT, Navisworks for Drawing and Modeling, supported by certifications and continuous trainings.

Airports & Stations



Project King AbdulAziz International Airport
Country Saudi Arabia
BUA 300,000 m²
Contractor SBG/Vision
Scope M&E Shop Drawings (PTB Zone D & Data Center) & Power Discrimination Study



Project Middle Euphrates International Airport
Country Iraq
BUA 200,000 m²
Consultant ADPi
Scope M&E Load Centers & PTB Infrastructure Detailed Design



Project Abha Airport
Country Saudi Arabia
BUA 87,000 m²
Architect Edge Arch
Scope M&E Detailed Design PTB & Parking (REVIT)



Project KKIA Airport
Country Saudi Arabia
Contractor Almabani
Scope M&E Design & Shop Drawings (Airfield Ground Lighting Stations)



Project ATCL Yacht Club & Marina
Country Lebanon
Area 250 Berths and 77 Yachts Pedestals
Client Automobile et Touring Club du Liban
Scope Quay and Sports Fields Area Lighting M&E Infrastructure Detailed Design



Project JAX Multistory Car Parks
Country Saudi Arabia
BUA 10,650 m²
Architect Edge Arch
Scope M&E Detailed Design (REVIT Works)



Project Riyadh Metro Stations, Line 5 & 6
Country Saudi Arabia
Contractor FCC Industrial
Scope M&E Shop Drawings & As-Built Drawings (REVIT Works for Technical Rooms)



Project Riyadh Metro Park & Ride
Country Saudi Arabia
Quantity 3 Stations
Contractor Freyssinet Saudi Arabia
Scope M&E Shop Drawings

Civic & Religious



Project **Baghdad Council of Ministers**
Country Iraq
BUA 178,000 m²
Architect CAP
Scope M&E Detailed Design & Follow Up



Project **UYO Conventional Center**
Country Nigeria
BUA 14,500 m²
Contractor ITB Nigeria
Scope Elec Detailed Design & Follow Up



Project **Samawah Cultural Center**
Country Iraq
BUA 12,000 m²
Architect CAP
Scope M&E Detailed Design



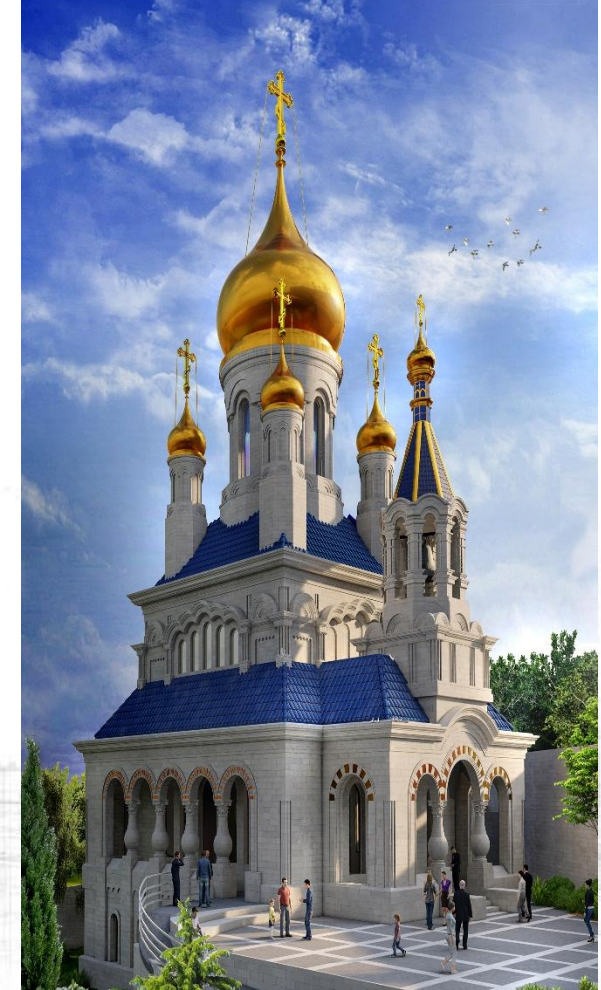
Project **King Abdullah International Gardens**
Country Saudi Arabia
Area 2,400,000 m²
Contractor ZHBC
Scope M&E Infrastructure and Buildings
Detailed Designs & Shop Drawings
REVIT for Energy Center



Project **Cité Marie**
Country Lebanon
BUA 31,500 m²
Client A Toi Marie Association
Scope M&E Design



Project **GACA Headquarters**
Country Saudi Arabia
BUA 108,000 m²
Contractor BinDayel
Scope M&E Detailed Design
REVIT Works for Tech Rooms



Project **Our Lady of Joy**
Country Lebanon
BUA 1,400 m²
Client Christian Culture & Heritage Foundation
Scope M&E Value Engineering



Project **Princess Noura University**
Country Saudi Arabia
BUA 80,000 m²
Contractor Saudi Oger
Scope Elec Shop Drawings (2 Faculties)



Project **Al Furat TV Broadcasting Station**
Country Iraq
BUA 10,000 m²
Architect CAP
Scope Elec Detailed Design



Project **Happy Childhood Village**
Country Tanzania
Land 232,000 m²
Architect ARK
Scope Elec Detailed Design



Project **Bayelsa Auditorium**
Country Nigeria
BUA 5,000 m²
Architect ARK
Scope M& E Detailed Design



Project **American International School**
Country Saudi Arabia
BUA 32,000 m²
Client Olayan Financing Company
Scope M&E Detailed Design



Project **Ibn Sina Hospital**
Country Iraq
BUA 79,000 m²
Architect CAP
Scope Elec Schematic Design



Project **PNU Medical Hospital**
Country Saudi Arabia
BUA 120,000 m²
Contractor Al Fayhaa / Oger
Scope Elec Shop Drawings



Project **Baghdad Teaching Hospital**
Country Iraq
BUA 59,000 m²
Architect CAP
Scope Elec Detailed Design



Project **CMRC Hospital I**
Country Saudi Arabia
BUA 6,525 m²
Client Cambridge Medical & Rehab Center
Scope M&E Value Engineering
M&E Redesign & Supervision



Project **CMRC Hospital II**
Country Saudi Arabia
BUA 11,450 m²
Client Richmond
Scope M&E Detailed Design (REVIT Works)



Project **Savior Medical Center**
Country Lebanon
BUA 49,300 m²
Architect Plan R
Scope Elec Schematic Design



Project **Shura Island - Hotel West #1**
Country Saudi Arabia
BUA 22,500 m²
Client First Gulf Company
Scope M&E Shop Drawings (REVIT Works)



Project **King Abdullah WAQF Tower Hotel**
Country Saudi Arabia
BUA 100,000 m²
Client Saudi Oger
Scope Elec Shop Drawings



Project **VOCO - IHG Hotel (234 Keys)**
Country Saudi Arabia
BUA 44,000 m²
Client Richmond
Scope M&E Detailed Design (REVIT Works)



Project **Sheraton Lagos Hotel**
Country Nigeria
BUA 22,200 m²
Contractor ITB Nigeria
Scope Elec Detailed Design



Project **MAAD Complex Hospitality Towers**
Country Saudi Arabia
BUA 225,700 m²
Client MAAD/Al-Jamed
Scope M&E Design & Shop Drawings
REVIT Works (Tech Floors)



Project **Hotel Complex Oasis (1001 Keys)**
Country Cuba
BUA 128,500 m²
Contractor Bouygues Bâtiment
Scope M&E Schematic Design (REVIT)



Project **Roca del Este Hotel I**
Country Cuba
BUA 47,000 m²
Contractor Bouygues Bâtiment
Scope M&E Schematic Design



Project **Hampton S7 Hotel**
Country Saudi Arabia
BUA 18,500 m²
Client MAEC
Scope M&E Detailed Design (REVIT Works)



Project **Vertical Indoor Farms**
Country Canada
BUA 15,700 m² + 15,900 m²
Contractor Kolostat
Scope Mech Construction & Fabrication Drawings (REVIT)



Project **New Jet Propulsion Center**
Country Saudi Arabia
BUA 60,000 m²
Contractor Freyssinet Saudi Arabia
Scope M&E Shop Drawings



Project **Eni Center of Excellence**
Country Congo
BUA 10,200 m²
Contractor Zerock
Scope Elec Shop Drawings



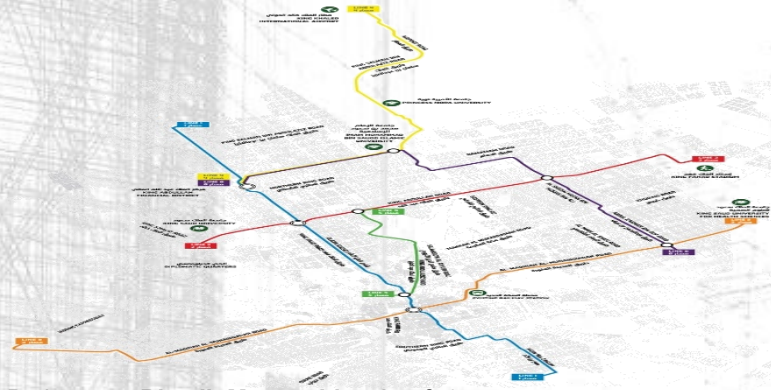
Project **Algorithm Pharmaceuticals**
Country Lebanon
BUA 15,000 m²
Contractor Betabat
Scope M&E Shop Drawings



Project **Industrial Biofuel Agri Hub**
Country Congo
BUA 6,800 m²
Contractor Zerock
Scope Electrical Detail Design



Project **Cervecería Cubana**
Country Latin America
BUA 15,000 m²
Client Bouygues Construction
Scope M&E Design, Shop Drawings & As-Built Drawings



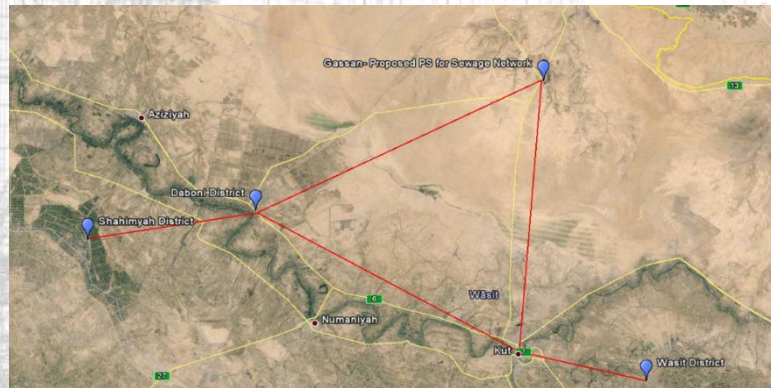
Project **Riyadh Metro – Line 4, 5 & 6**
Country Saudi Arabia
Contractor FAST / Samsung
Scope Road Safety Study



Project **Jannat Baghdad Development**
Country Iraq
Area 28 clusters of 720 Residences,
42 Schools & 20 Mixed-Use Buildings
Architect CAP
Scope M&E Infrastructure Schematic Design



Project **Saafat El Basra City**
Country Iraq
Area 600,000 m²
Client Al Saqr Al Jareh
Scope M&E Infrastructure Detailed Design
Architecture and M&E Buildings Detailed Design



Project **Wasit Water & Sanitation Stations**
Country Iraq
Districts Wasit, Daboni, Gassan & Shahimiyah
Consultant CAP
Scope Drainage & Rainwater Infrastructure



Project **Diriyah Square – Mega Car Park**
Country Saudi Arabia
Area 330,000 m²
Contractor First Gulf Company
Scope MEP Shop Drawings (REVIT Works)



Project **Venan Housing Project**
Country Saudi Arabia
BUA 450,000 m²
Architect Edge Arch
Scope M&E Infrastructure & Buildings Detailed Design

Mixed Use



Project EKO Tower II
(Residential & Medical Centre)
Country Nigeria
BUA 35,500 m²
Client ITB Nigeria
Scope Elec Detailed Design & Follow up



Project Azuri Peninsula
(Residential & Offices)
Country Nigeria
BUA 123,000 m²
Client ITB Nigeria
Scope Electrical Value Engineering & Shop Drawings



Project Karbala Oasis Development
(Mall & Hotel)
Country Iraq
BUA 200,000 m²
Architect CAP
Scope M&E Detailed Design



Project Palm Towers
(Residential & Offices)
Country Côte d'Ivoire
BUA 11,400 m²
Architect AOT
Scope M&E Detailed Design / M&E Shop Drawings



Project Lawyers & Notaries Tower
(Hotel & Offices)
Country Iraq
BUA 200,000 m²
Client NGC
Scope M&E Detailed Design

Offices



Project **Crystal Tower**
Country Nigeria
BUA 18,000 m²
Client RMCK Construction
Scope M&E Detailed Design



Project **Garden Heights**
Country Nigeria
BUA 20,000 m²
Architect Archico Consultants
Scope M&E Detailed Design



Project **EKO Tower I
(TOTAL Headquarters)**
Country Nigeria
BUA 44,000 m²
Client ITB Nigeria
Scope Elec Detailed Design & Follow up



Project **Kingsway Tower
(Microsoft & Facebook Offices)**
Country Nigeria
BUA 14,830 m²
Contractor ITB Nigeria
Scope Elec Shop Drawings



Project **World Bank**
Country Nigeria
BUA 4,200 m²
Client ITB Nigeria
Scope Elec Shop Drawings



Project **Otunba Offices**
Country Nigeria
BUA 8,000 m²
Architect Domaine Public Architects
Scope M&E Detailed Design



Project **The Carnelian**
Country Nigeria
BUA 33,000 m²
Architect Archico Consultants
Scope M&E Detailed Design



Project **Riyadh Power Plant PP10**
Country Saudi Arabia
Size 2,500 MW Power Generation
Contractor Arabian Bemco Contracting
Scope M&E Detailed Design (Industrial Bldgs)
Outdoor & Road Lighting



Project **Al-Qassim PP Extension #2**
Country Saudi Arabia
Size 560 MW Power Generation
Contractor Arabian Bemco Contracting
Scope M&E Design (Industrial Bldgs)
Outdoor & Road Lighting



Project **Al-Qassim PP Extension #3**
Country Saudi Arabia
Size 560 MW Power Generation
Contractor Arabian Bemco Contracting
Scope M&E Design (Industrial Bldgs)
Outdoor & Road Lighting



Project **King Khaled International Airport AGL Substations**
Country Saudi Arabia
Size 4 MW Power Generation
Contractor Almamani
Scope Architecture and M&E Shop Drawings



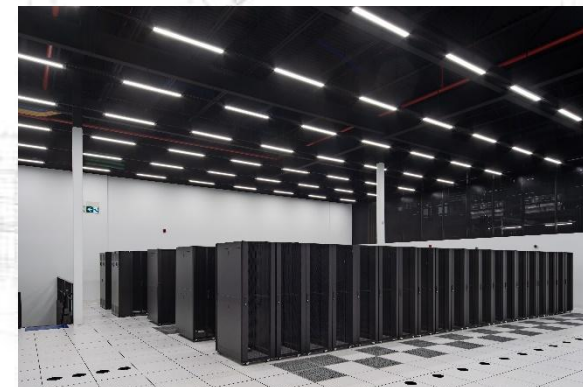
Project **KAIA Data Centers 1&2**
Country Saudi Arabia
Size 11 MW
Contractor SBG/Vision
Scope M&E Shop Drawings



Project **Total Headquarters Data Center**
Country Nigeria
Size 500 KW
Contractor ITB Nigeria
Scope Elec Detailed Design



Project **Africa Data Centers**
Country Nigeria
Size 750 KW
Contractor ITB Nigeria
Scope Electrical Shop Drawings

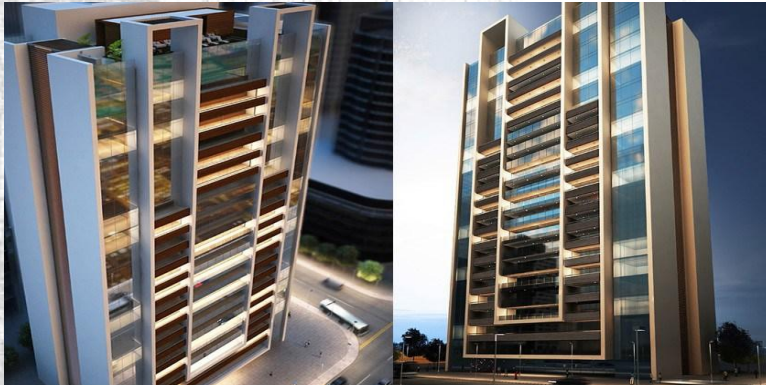


Project **eStruxture Data Center**
Country Canada
Size 7x230 Ton Cooling Load
Contractor Kolostat
Scope Mechanical Shop Drawings

Residential



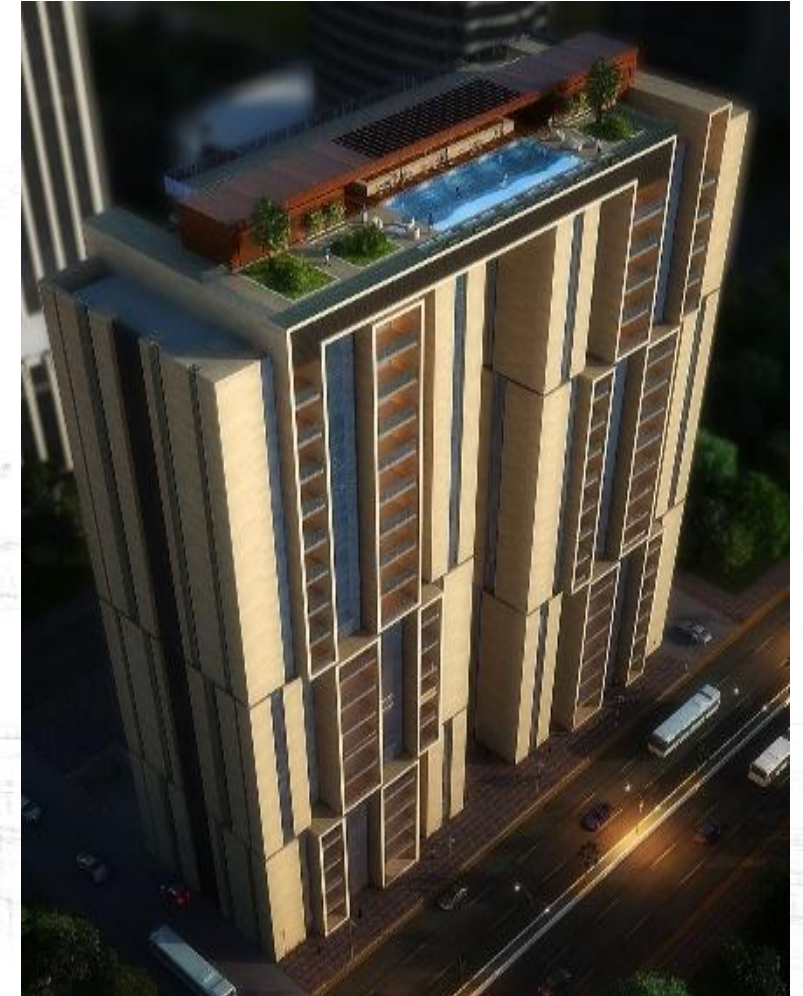
Project **Meydan I Staff Accommodation**
Country UAE
BUA 62,500 m²
Client EMW
Scope M&E Shop Drawings



Project **Sky Tower**
Country Nigeria
BUA 32,700 m²
Client NGC
Scope M&E Detailed Design



Project **Kuramo Beach Residence**
Country Nigeria
BUA 71,500 m²
Client ITB Nigeria
Scope Elec Shop Drawings
(REVIT for Elec Rooms)



Project **Twin Tower**
Country Nigeria
BUA 41,500 m²
Client RMCK
Scope M&E Detailed Design



Project **Zouhour Baghdad Mall**
Country Iraq
BUA 150,000 m²
Architect CAP
Scope M&E Detailed Design



Project **Kirkuk Mall**
Country Iraq
BUA 62,500 m²
Architect CAP
Scope M&E Detailed Design



Project **Baghdad Mall**
Country Iraq
BUA 750,000 m²
Client NGC
Scope Elec Design Review



Project **Marchés Urbains et Régionaux**
Country Benin
BUA 9 Markets Districts
Contractor ITB Nigeria
Scope M&E Shop Drawings



Project **Kirkuk Landmark Mall**
Country Iraq
BUA 55,500 m²
Architect CAP
Scope M&E Detailed Design



Project **Erbil City Center**
Country Iraq
BUA 637,750 m²
Client NGC
Scope Elec Detailed Design



Project **MAG 5 Boulevard**
Country UAE
BUA 140,000 m²
Owner MAG
Scope M&E Value Engineering



Project **MBL Residence**
Country UAE
BUA 70,600 m²
Owner MAG
Scope M&E Value Engineering

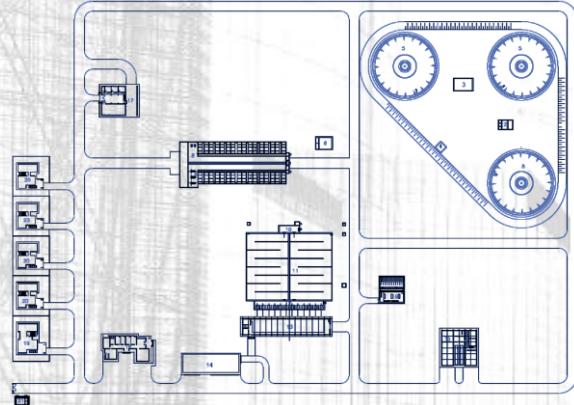


Project **Entissar Tower
(Residential & Luxury Hotel)**
Country UAE
BUA 270,500 m² (570 m height)
Consultant WSP / ae7
Scope M&E Value Engineering



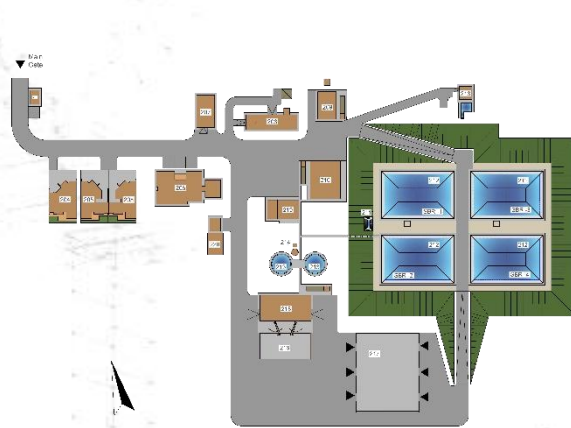
Project **Viceroy Tower**
Country UAE
BUA 140,000 m²
Consultant Atkins
Scope M&E Value Engineering

Water Treatment Plants

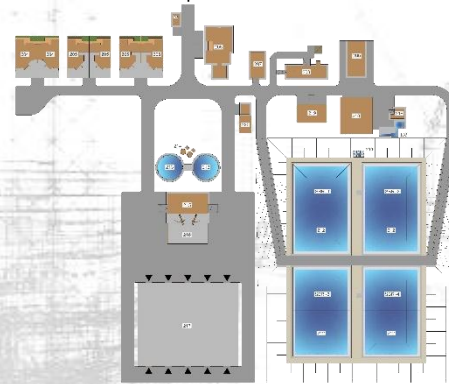


Project
Country
Brief
Scope

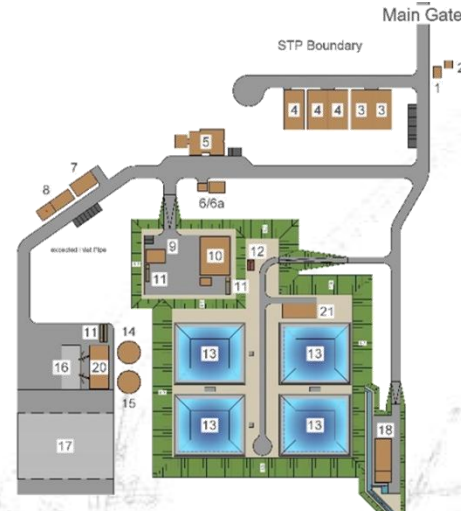
GOVERNORATE OF QAYARA PROJECT
Iraq
River Water Treatment Plant
Area: 0.13 km² (13 hectare)
Flow: 5500 LPS
M&E Detailed Design
Network infrastructure
Pumping Stations



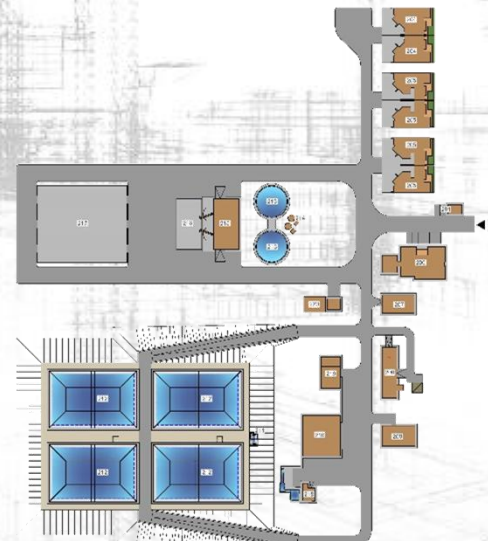
GHASSAN DISTRICT



WASSIT DISTRICT



SHAHIMIYAH DISTRICT



DABONI DISTRICT

Project
Country
Brief

GOVERNORATE OF WASIT SEWERAGE PROJECT
Iraq
4 Sewage Treatment Plants using SBR technology
for 4 districts in the Governorate of Wasit

Gassan

Area: 1.62 km² (162 hectare)
Population: 28,100 persons
Pump Stations: 2x3200 LPS each
Sewage Treatment Plant: 278 LPS

Wassit

Area: 4.15 km² (415 hectare)
Population: 72,000 persons
Pump Stations: 3x3200 LPS each, 1x6400LPS
Sewage Treatment Plant: 648 LPS

Shahimiyah

Area: 1.33 km² (130 hectare)
Population: 23,100 persons
Pump Stations: 2x3200 LPS each
Sewage Treatment Plant: 278 LPS

Daboni

Area: 2.72 km² (293 hectare)
Population: 47,183 persons
Pump Stations: 3x(1600 LPS, 3200 LPS, 6400 LPS)
Sewage Treatment Plant: 509 LPS

Scope

M&E Detailed Design
Network infrastructure
Pumping Stations
Sewage Treatment Plants



International Fair of Tripoli, Lebanon by Oscar Niemeyer – Unesco World Heritage

emdc
G r o u p

Al Moudir Center,
Jal el Dib, El Metn, Lebanon
+961 4 710 253 | +961 4 719 253

Dubai Silicon Oasis, Dubai Digital Park,
Building A2, Dubai, United Arab Emirates
+971 54 421 1769